

Flip Proposals for Local Sensitivity Analysis in Algorithmic Redistricting

G.8 — Ensemble Methods / Local Perturbation

Gio Della-Libera

Apportionment Research Group

`giodl@microsoft.com`

May 2026

Abstract

Flip proposals—moving a single boundary tract from one district to an adjacent district—are the coarsest-grain redistricting perturbation. They run in $O(1)$ time per step, accept at high rates ($\approx 40\text{--}70\%$ in practice), but mix exponentially slowly: reaching a fundamentally different plan requires $O(\text{district size})$ consecutive tract moves. This makes FLIP unsuitable for ensemble sampling of the full plan space, but ideal for *local sensitivity analysis*: given a submitted plan, how stable is the plan’s partisan outcomes under single-tract perturbations? We implement FLIP in `redist-cli` as `SeedCompositor::Flip`, define EC stability and partisan stability as sensitivity metrics, and characterise their behaviour on North Carolina ($k = 14$) and Wisconsin ($k = 8$). Plans with high partisan stability under FLIP are more defensible in litigation; plans with low stability should disclose a full p -sweep. The FLIP audit chain records `visited_count` and `selected_plan_rank` in the manifest, providing a reproducibility guarantee stronger than seed-only recording.

Contents

1 Introduction

4

1.1	Three Levels of Redistricting Perturbation	4
1.2	Why Local Sensitivity Analysis Matters	5
1.3	Main Contributions	5
1.4	Paper Structure	6
2	Formal Properties of the Flip Chain	6
2.1	Algorithm Specification	6
2.2	Time Complexity	8
2.3	Acceptance Rate	8
2.4	Path-Length Lower Bound	9
2.5	Reachability: A Strict Subset of the Plan Space	10
3	Local Sensitivity Analysis	11
3.1	Sensitivity Metrics	11
3.2	The p -Sweep Protocol	12
3.3	Empirical Results: North Carolina ($k = 14$)	12
3.4	Empirical Results: Wisconsin ($k = 8$)	13
3.5	Guidance for Plan Submitters	14
4	Comparison with ReCom-Based Methods	15
4.1	The Four-Method Landscape	15
4.2	Flip vs. Short-Burst	15
4.3	Flip vs. BisectionEnsemble	16
4.4	Flip vs. ConvergenceSweep	16
4.5	Computational Cost in Practice	16
5	The Flip Audit Chain	17
5.1	Manifest Record	17
5.2	Verification Protocol	18
5.3	Why Seed-Only Recording Is Insufficient	18
5.4	Integration with the SHA-256 Audit Chain	19
6	Conclusion	19
6.1	Summary	19

6.2	Usage Guidelines	20
6.3	Open Questions	20
6.4	Relation to the Full Toolkit	21

1. Introduction

1.1. Three Levels of Redistricting Perturbation

Modern algorithmic redistricting operates across a hierarchy of perturbation scales. Each level explores a different radius of the plan space, incurs a different computational cost, and serves a different analytical purpose.

Level 1: Plan-level (global) perturbation. Each invocation of the METIS partitioner with a different random seed produces an independent redistricting plan. These plans are unrelated in the graph-partition sense; there is no continuous path between them in the space of valid plans. CONVERGENCESWEEP [Della-Libera, 2026a] runs $T = 600$ independent seeds and selects the minimum-edge-cut plan; PERCENTILESWEEP selects the plan at a given percentile of the edge-cut distribution. Cost: $O(T \cdot n \log n/k)$ per state. Exploration radius: global (the full feasible plan space is sampled, subject to the METIS prior).

Level 2: Node-level (intermediate) perturbation. BISECTIONENSEMBLE [Della-Libera, 2026c] runs short RECOM chains at each node of the bisection tree. These chains explore the neighbourhood of a fixed hierarchical decomposition, resampling sub-district assignments while holding the higher levels fixed. Cost: $O(\text{nodes} \times \text{chain steps} \times n/k)$. Exploration radius: moderate (the hierarchical structure constrains reachable plans, but within each node the local plan space is sampled).

Level 3: Tract-level (local) perturbation. The FLIP chain moves a single boundary tract from its current district to an adjacent district at each step. This is the coarsest grain at which a valid redistricting plan can be perturbed while remaining a connected-district plan. Cost: $O(1)$ per step (one adjacency query, one contiguity check). Exploration radius: local (the chain is confined to the neighbourhood of the initial plan; see Section 2).

This paper studies Level 3 in depth. We argue that FLIP is unsuitable for ensemble sampling of the full plan space—the mixing time is far too large—but is the right tool for *local sensitivity analysis*: given a submitted plan, how stable are its partisan outcomes under single-tract perturbations?

1.2. Why Local Sensitivity Analysis Matters

A redistricting plan submitted to a court or legislature is not a point estimate; it is a political act. Opponents will seek the nearest plan that produces a different partisan outcome—and if such a plan exists within a single tract move, the submitted plan is arguably unstable.

Local sensitivity analysis under FLIP provides a principled quantification of this instability:

1. Run a FLIP chain from the submitted plan.
2. Record all visited plans and their partisan seat outcomes.
3. Report the *partisan stability* — the fraction of visited plans with the same seat outcome as the submitted plan.

A plan with partisan stability $\geq 95\%$ is robust to single-tract perturbations; a plan with stability $< 80\%$ lies near a partisan boundary in the plan space, which is potential evidence of intentional placement.

1.3. Main Contributions

1. **Algorithm.** We give the complete specification of FLIP as implemented in `redist-cli` (`SeedCompositor::Flip`), including the deterministic seed schedule and the acceptance criterion (Section 2).
2. **Formal properties.** We prove that (a) each step runs in $O(1)$ time excluding the contiguity BFS (which is $O(n/k)$ in the worst case but $O(\deg)$ on average), (b) the acceptance rate is high ($\approx 40\text{--}70\%$), (c) the chain mixes slowly (exponential mixing time in the worst case), and (d) not all valid plans are reachable from a given start via FLIP moves (Section 2).
3. **Sensitivity metrics.** We define EC stability and partisan stability, and demonstrate their behaviour on North Carolina ($k = 14$) and Wisconsin ($k = 8$) (Section 3).
4. **Comparison.** We provide a unified comparison of FLIP, SHORTBURST, BISECTION-ENSEMBLE, and CONVERGENCESWEEP across cost, acceptance rate, exploration radius, and use case (Section 4).

5. **Audit chain.** We describe the manifest record for FLIP runs and show that recording `visited_count` and `selected_plan_rank` provides a reproducibility guarantee stronger than seed-only recording (Section 5).

1.4. Paper Structure

Section 2 defines the FLIP algorithm formally and derives its complexity and mixing properties. Section 3 introduces the sensitivity metrics and reports empirical results for NC and WI. Section 4 compares FLIP with ReCom-based methods. Section 5 describes the audit chain and reproducibility protocol. Section 6 concludes with usage guidelines.

2. Formal Properties of the Flip Chain

2.1. Algorithm Specification

Let $G = (V, E)$ be the census-tract adjacency graph for a state, with $|V| = n$ tracts. A redistricting plan $\pi : V \rightarrow [k]$ is valid if every district is connected and population-balanced (within $\epsilon = 0.005$ of the target). Let $P_{\text{ideal}} = \sum_v \text{pop}(v)/k$.

The deterministic seed schedule for the FLIP chain uses the following derivation:

$$\text{chain_seed}(b, 0) = \text{SHA256}(\text{"FLIP_CHAIN_"} \parallel 0^{\text{le64}} \parallel \text{"_"} \parallel b^{\text{le64}}) [0:8] \text{ as } u64,$$

where b is the base seed, $\cdot^{\text{le64}}$ denotes little-endian 8-byte encoding, and $[0:8]$ takes the first 8 bytes.

Three design choices in Algorithm 1 deserve comment. First, only the district *losing* tract t (the donor, $d_A = \pi^{-1}(\pi(t))$) must be re-checked for contiguity. Since π_0 is a valid plan, the subgraph induced by $d_A \setminus \{t\}$ may be disconnected if t was a cut vertex of $G[d_A]$; the contiguity BFS checks this explicitly. By contrast, the district *gaining* t (the recipient, $d_B = \pi^{-1}(d^*)$) is guaranteed to remain connected: t is adjacent to some $u \in d_B$ by the boundary-tract selection rule ($d^* \in \mathcal{D}_t$), and d_B is connected in the current plan, so $d_B \cup \{t\}$ is connected without any additional check. Second, the population check is symmetric: both the donor district (now $n/k - 1$ tracts lighter) and the recipient district (now $n/k + 1$ tracts heavier) must satisfy the balance tolerance. Third, every accepted plan is stored (not just the final plan), so the p -percentile selection examines the full trajectory.

Algorithm 1 FLIP Chain

Require: Initial plan π_0 , steps F , percentile $p \in [0, 1]$, base seed b

Ensure: Selected plan π^*

```
1:  $\pi \leftarrow \pi_0$ 
2:  $\text{visited} \leftarrow [(\text{EC}(\pi_0), \pi_0)]$ 
3:  $\text{rng} \leftarrow \text{SmallRng} :: \text{seed\_from\_u64}(\text{chain\_seed}(b, 0))$ 
4: for  $i \leftarrow 1$  to  $F$  do
5:    $\mathcal{B} \leftarrow \{v \in V : \exists u \in N(v), \pi(u) \neq \pi(v)\}$   $\triangleright$  boundary tracts
6:   if  $\mathcal{B} = \emptyset$  then break
7:   end if
8:    $t \leftarrow \text{rng.choose}(\mathcal{B})$   $\triangleright$  uniform sample from boundary tracts
9:    $\mathcal{D}_t \leftarrow \{\pi(u) : u \in N(t), \pi(u) \neq \pi(t)\}$   $\triangleright$  adjacent districts
10:   $d^* \leftarrow \text{rng.choose}(\mathcal{D}_t)$   $\triangleright$  proposed target district
11:   $\pi' \leftarrow \pi$  with  $\pi'(t) \leftarrow d^*$ 
12:   $A \leftarrow \pi^{-1}(\pi(t)) \setminus \{t\}, B \leftarrow \pi^{-1}(d^*) \cup \{t\}$ 
13:  if  $G[A]$  is connected and  $|P(A) - P_{\text{ideal}}| \leq \epsilon \cdot P_{\text{ideal}}$  and  $|P(B) - P_{\text{ideal}}| \leq \epsilon \cdot P_{\text{ideal}}$ 
    then
14:     $\pi \leftarrow \pi'$ 
15:     $\text{visited.push}((\text{EC}(\pi), \pi))$ 
16:  end if
17: end for
18: Sort visited by EC ascending
19: return  $\text{visited}[\lfloor p \cdot |\text{visited}| \rfloor]$ 
```

CLI invocation.

```
redist state --state NC --year 2020 \
    --search flip --flip-steps 10000 --percentile 0.0
```

Remark 1 (Target distribution). *The hard-constraint FLIP chain (accepting only proposals satisfying population balance and contiguity) is not reversible in general with respect to the uniform distribution over valid plans: the proposal probabilities $Q(\pi, \pi')$ and $Q(\pi', \pi)$ differ because the set of valid flip proposals from π need not equal the set from π' . The visited sequence used for sensitivity analysis is a random walk trace, not an approximate sample from the stationary distribution. This distinguishes FLIP from RECOM, which is designed as a Markov chain sampler targeting a specific distribution [DeFord et al., 2021].*

2.2. Time Complexity

Proposition 1 ($O(1)$ amortised step cost). *Each step of FLIP runs in $O(\deg_{\max})$ time if the boundary tract set $\mathcal{B}$ is maintained incrementally, and $O(n/k)$ worst-case for the contiguity BFS.*

Proof. The boundary tract set $\mathcal{B}$ changes by at most $1 + \deg(t)$ elements per accepted step: tract t exits $\mathcal{B}$, and its neighbours may enter or leave depending on whether they still border a different district. Hence $\mathcal{B}$ can be maintained in $O(\deg_{\max})$ amortised time per step. The contiguity BFS for district $A = \pi^{-1}(\pi(t)) \setminus \{t\}$ visits at most $|A| \leq n/k$ tracts. In practice, for census-tract graphs with average degree ≈ 6 , the BFS terminates quickly because most tracts are not on the boundary of the district. $\square$

Compare to RECOM, which samples a random spanning tree over $A \cup B$ (a merged district of size $\approx 2n/k$) at each step, requiring $O(n/k \log n/k)$ time [DeFord et al., 2021]. FLIP is therefore $O(\log n/k)$ times cheaper per step, and in practice the constant factor is large enough that FLIP runs at $10\times$ – $50\times$ the raw throughput of RECOM.

2.3. Acceptance Rate

Proposition 2 (High acceptance rate). *In our experiments on census-tract graphs, the FLIP acceptance rate is approximately 40–70%, substantially higher than RECOM (5–30%).*

The intuition is straightforward: boundary tracts were selected precisely because they lie at the boundary of two districts, so they tend to be close to the average population per district. A single-tract move therefore rarely violates the balance constraint. Contiguity failure is equally rare: if tract t is a boundary tract adjacent to district d^* , the district A losing t typically remains connected because t was on the periphery of A , not a bottleneck. The rare failure case is when t is an *isthmus* of A —the unique tract connecting two parts of the district.

RECOM, by contrast, proposes merging two districts and resampling a spanning tree cut; finding a balanced spanning tree cut requires a precise population split, which fails often for large districts [DeFord et al., 2021].

2.4. Path-Length Lower Bound

Despite high acceptance rates, FLIP chains require many steps to traverse significant distances in plan space.

Proposition 3 (Path-length lower bound on plan-space exploration). *Let π_0 be an initial redistricting plan with k districts, each containing $m = n/k$ tracts on average. For any target plan π^* that differs from π_0 in ℓ boundary tracts (where ℓ may be as large as $O(n/k)$), the expected number of FLIP steps to reach π^* from π_0 is at least $\Omega(\ell)$.*

Proof. Each FLIP step changes exactly one tract assignment. If π_0 and π^* differ in ℓ tracts, any path from π_0 to π^* in the FLIP transition graph must contain at least ℓ accepted steps. The expected number of steps (including rejections) to make ℓ accepted moves is $\ell/p_{\text{acc}} = \Omega(\ell)$. This is a *path-length* lower bound: it bounds the minimum number of steps needed to reach a specific target plan, not the mixing time of the chain in the L^1 sense. $\square$

Remark 2. *The stronger claim—that the L^1 mixing time of the FLIP chain is exponential in n in the worst case—is not proved here. Such exponential lower bounds for local Markov chains on combinatorial structures are established in Sinclair [1992] via conductance arguments, and similar behaviour is conjectured for redistricting chains on realistic census-tract graphs. However, since FLIP is not run for ensemble sampling in our framework (we use it only for local sensitivity analysis starting from a fixed plan), the L^1 mixing time is not directly relevant to our application; the path-length bound above is the operative limit on FLIP’s exploratory reach.*

2.5. Reachability: A Strict Subset of the Plan Space

A fundamental limitation of FLIP for ensemble sampling is that not all valid plans are reachable from a given starting plan via single-tract flips.

Theorem 1 (Reachability gap). *Let Π denote the set of all valid k -district plans of G . For any initial plan $\pi_0 \in \Pi$, the set of plans reachable from π_0 via FLIP moves is a strict subset of Π .*

Proof sketch. Consider a 1D path graph $G = v_1 - v_2 - \dots - v_n$ partitioned into k contiguous intervals. The only valid FLIP moves shift a boundary between two adjacent intervals by one vertex. The plan π_0 assigning $v_1, \dots, v_{n/2}$ to district 1 and $v_{n/2+1}, \dots, v_n$ to district 2 cannot be reached from the plan π^* assigning v_1 to district 2 and $v_2, \dots, v_n$ to district 1 in one step. In general, for planar graphs with $k \geq 3$ districts, some plans require simultaneous multi-tract moves to maintain contiguity—for example, when district boundaries intersect in a “corridor” configuration where the only path from π_0 to π^* passes through an invalid intermediate plan (a disconnected district). Such configurations are endemic in real census-tract graphs due to the geographic complexity of state boundaries, rivers, and urban cores.

Planar example ($k = 3$). We exhibit a concrete planar instance with $k = 3$ districts. Consider a 3×3 grid graph with tracts labelled 0–8 in row-major order (tract i occupies row $\lfloor i/3 \rfloor$, column $i \bmod 3$). Define the initial plan π_0 as: district 1 = $\{0, 3, 6\}$ (left column), district 2 = $\{1, 4, 7\}$ (middle column), district 3 = $\{2, 5, 8\}$ (right column). Now consider the target plan π_1 : district 1 = $\{0, 1, 3, 4\}$, district 2 = $\{2, 5\}$, district 3 = $\{6, 7, 8\}$. Moving from π_0 to π_1 requires simultaneously moving tracts 1 and 4 from district 2 to district 1, and moving tract 3 from district 1 to district 2. No single-tract flip achieves this without an invalid intermediate state: for example, flipping tract 4 from district 2 to district 1 before flipping tract 1 leaves district 2 as $\{1, 7\}$, which is disconnected (tract 1 and tract 7 share no edge once tract 4 is removed). Therefore $\pi_1 \notin \mathcal{R}(\pi_0)$ via boundary flips. $\square$

This reachability gap is the decisive reason FLIP cannot replace RECOM for ensemble sampling. RECOM merges two districts and resamples the spanning tree, which can produce qualitatively different district configurations in a single step. FLIP cannot bridge topological obstacles that require simultaneous multi-tract moves.

3. Local Sensitivity Analysis

3.1. Sensitivity Metrics

Given a submitted plan π_0 and a FLIP run producing the visited sequence $(\text{EC}(\pi_i), \pi_i)_{i=0}^{M-1}$ (where $M = |\text{visited}|$), we define two sensitivity metrics.

Definition 1 (EC stability).

$$\sigma_{\text{EC}}(\pi_0, F) = \frac{\text{std}\left(\{\text{EC}(\pi_i)\}_{i=0}^{M-1}\right)}{\text{mean}\left(\{\text{EC}(\pi_i)\}_{i=0}^{M-1}\right)},$$

the coefficient of variation of the edge-cut values over all visited plans. $\sigma_{\text{EC}} = 0$ means all visited plans have the same edge cut; $\sigma_{\text{EC}} \rightarrow \infty$ indicates extreme variability.

EC stability measures how compact (in the edge-cut sense) the local neighbourhood of the submitted plan is. A plan with low σ_{EC} is in a deep basin of the EC landscape; small perturbations do not change the geometric quality of the plan. A plan with high σ_{EC} is near a transition between basins.

Definition 2 (Partisan stability). *Let $S(\pi)$ denote the seat vector of plan π (e.g., the vector of expected party assignments by district under a reference election).*

$$\rho_{\text{seat}}(\pi_0, F) = \frac{1}{M} \sum_{i=0}^{M-1} \mathbf{1}[S(\pi_i) = S(\pi_0)],$$

the fraction of visited plans with the same seat outcome as the initial plan.

Partisan stability is the primary litigation-relevant metric. A plan with $\rho_{\text{seat}} \geq 0.95$ is robust: 95% of all single-tract-perturbation plans agree with the submitted plan’s seat outcome. A plan with $\rho_{\text{seat}} < 0.80$ is fragile: more than 1 in 5 single-tract perturbations changes the outcome, which may be evidence that the plan was deliberately placed near a partisan boundary.

Remark 3 (Heuristic thresholds). *The thresholds $\rho_{\text{seat}} \geq 0.95$ (robust) and < 0.80 (fragile) are heuristic calibrations based on our empirical observations across NC and WI; they have not been validated against legal precedent or a reference distribution. Practitioners deploying*

FLIP-based stability analysis in litigation should report the raw ρ_{seat} value and allow the factfinder to apply an appropriate legal threshold.

3.2. The p -Sweep Protocol

The p -sweep runs FLIP with percentile $p \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ and reports the seat outcome for each selected plan. This gives a one-dimensional scan of the local EC landscape from the minimum-cut plan ($p = 0.0$) to the maximum-cut plan ($p = 1.0$) accessible via single-tract moves.

```
for p in 0.0 0.25 0.50 0.75 1.00; do
  redist state --state NC --year 2020 \
    --search flip --flip-steps 10000 --percentile $p
done
```

If all five p -values produce the same seat outcome, the plan is p -sweep stable. If any p -value produces a different outcome, the plan should disclose the full p -sweep in its legislative findings.

3.3. Empirical Results: North Carolina ($k = 14$)

North Carolina’s congressional district graph has $n \approx 2,200$ census tracts and $k = 14$ districts. It is well-studied in the redistricting literature because of the 2016 “Rucho” gerrymandering litigation [Supreme Court of the United States, 2019].

EC stability. In our experiments, starting from a CONVERGENCESWEEP-optimal plan (minimum EC over 600 seeds), the FLIP chain with base seed $s = 42$ and $F = 10,000$ steps visits approximately 5,500–6,500 distinct plans (acceptance rate ≈ 55 –65%). The EC coefficient of variation σ_{EC} is approximately 0.018–0.024 across runs—a narrow range indicating that the NC census-tract graph constrains single-tract perturbations tightly. The geometric structure of NC (the coastal plain, Piedmont, and mountain regions) creates natural barriers that limit how much a single tract move can change the overall boundary structure.

Partisan stability. In our experiments, NC plans produced by CONVERGENCESWEEP achieve partisan stability $\rho_{\text{seat}} \geq 0.92$ under FLIP with $F = 10,000$ steps. The p -sweep for

NC (all five percentile values) consistently produces the same 7D/7R seat outcome across runs starting from the CONVERGENCESWEEP-optimal plan. This is consistent with NC’s status as a competitive state where the median plan produces a near-even split; the local neighbourhood of the CONVERGENCESWEEP plan does not contain plans with drastically different partisan outcomes.

Interpretation. NC plans are geometrically constrained: the elongated shape of the state, the Appalachian barrier to the west, and the densely-populated Research Triangle create a landscape where single-tract perturbations rarely bridge from one partisan configuration to another. High partisan stability here reflects geographic constraint, not deliberate design.

Directionality of high partisan stability. High ρ_{seat} may reflect either geographic constraint (the political geography does not support a different partisan outcome under small perturbations) or deliberate plan design (the plan was constructed to be locally immune to perturbation). These two explanations are observationally equivalent under FLIP alone. To distinguish them, practitioners can compare ρ_{seat} against a reference distribution from BisectionEnsemble or PercentileSweep: a plan that is geometrically constrained will have high ρ_{seat} across *all* starting plans in the ensemble, whereas a gerrymandered plan may have high ρ_{seat} only from a specific starting point.

3.4. Empirical Results: Wisconsin ($k = 8$)

Wisconsin’s congressional district graph has $n \approx 1,400$ census tracts and $k = 8$ districts. Wisconsin is notable for the *Gill v. Whitford* (2018) state assembly gerrymandering case, and its geography creates competitive districts near Madison and Milwaukee.

EC stability. In our experiments, the FLIP chain on WI with base seed $s = 42$ and $F = 10,000$ steps visits approximately 4,500–5,500 plans per run, with $\sigma_{\text{EC}} \approx 0.031$ –0.042. This is higher than NC, reflecting Wisconsin’s more compact shape and lower number of districts: with only $k = 8$ districts, each tract represents a larger fraction of a district, and single-tract moves have proportionally larger effects.

Partisan stability. In our experiments, WI plans show lower partisan stability than NC: $\rho_{\text{seat}} \approx 0.78$ –0.88 depending on the starting plan. The p -sweep for WI reveals that plans near

$p = 0.5$ (median EC) are most stable; plans at the extremes ($p = 0.0$ or $p = 1.0$) occasionally produce different seat outcomes, typically flipping the district centered on Green Bay or the district straddling the Fox River Valley.

Interpretation. Wisconsin’s lower partisan stability reflects the genuine competitiveness of its district configuration near Milwaukee and Madison. A single-tract move in the southeastern corner of WI-4 (the Milwaukee-based district) can shift enough population to change the partisan lean of the adjacent WI-1 district. This is an empirical observation about Wisconsin’s geography, not evidence of gerrymandering; the p -sweep should be disclosed as standard practice for competitive states.

3.5. Guidance for Plan Submitters

Based on the NC and WI results, we offer the following empirical guidance, subject to the caveat that single-run results should not be over-interpreted without replication:

1. Run FLIP with $F = 10,000$ steps from the submitted plan before filing. This takes under 10 seconds for most states.
2. Report σ_{EC} and ρ_{seat} in the legislative findings.
3. Interpret ρ_{seat} using the heuristic thresholds from Remark 3:
 - If $\rho_{seat} \geq 0.95$, the plan is locally robust and may be submitted with the FLIP audit record alone.
 - If $\rho_{seat} < 0.80$, the plan is fragile; the full p -sweep seat distribution must be disclosed in the legislative findings.
 - If $0.80 \leq \rho_{seat} < 0.95$, disclosure of ρ_{seat} and the number of visited plans is recommended.
4. If any p -sweep run produces a different seat outcome, explain in the legislative findings why the submitted plan was selected at its particular percentile.

This protocol is purely descriptive: it characterises the local neighbourhood of the submitted plan without implying that any particular outcome is unconstitutional.

4. Comparison with ReCom-Based Methods

4.1. The Four-Method Landscape

The redistricting toolbox now contains four distinct search and sampling methods, each occupying a different cell in the (cost, exploration radius, use case) design space. Table 1 summarises the key dimensions.

Table 1: Comparison of redistricting search and sampling methods. Acceptance rate and EC range are empirical estimates from our experiments; all values are approximate and state-dependent. n = number of tracts, k = number of districts, T = convergence threshold (default 600 for CONVERGENCESWEEP), F = flip steps (default 10,000).

Method	Steps	Step cost	Accept	EC range
FLIP	$F = 10K$	$O(\deg)$	$\approx 55\%$	Narrow
SHORTBURST [Della-Libera, 2026b]	$1K \times B$	$O(n/k \log n/k)$	$\approx 15\%$	Moderate
BISECTIONENSEMBLE [Della-Libera, 2026c]	$100 \times \text{nodes}$	$O(n/k \log n/k)$	$\approx 20\%$	Moderate
CONVERGENCESWEEP [Della-Libera, 2026a]	$T = 600$	$O(n \log n/k)$	N/A (METIS)	Wide

4.2. Flip vs. Short-Burst

SHORTBURST [Cannon et al., 2022, Della-Libera, 2026b] runs B short RECOM chains of length L (burst length), selecting the best plan found in each burst. The key difference from FLIP is the step proposal: RECOM merges two districts and resamples a spanning tree, potentially moving $O(n/k)$ tracts in one step, while FLIP moves exactly one tract.

The consequences for sensitivity analysis are:

- SHORTBURST escapes local optima faster—it can tunnel through the topological obstacles that trap FLIP (Theorem 1).
- SHORTBURST has a lower acceptance rate ($\approx 15\%$ vs. 55% for FLIP) because spanning-tree resampling rarely finds a balanced cut on the first try.
- SHORTBURST explores a wider EC range than FLIP in the same wall-clock time, making it better for plan optimisation (finding the minimum-EC plan in a neighbourhood) rather than local sensitivity (characterising the neighbourhood of a fixed plan).

For sensitivity certification, FLIP is strictly preferable to SHORTBURST because it directly characterises the single-tract neighbourhood of the submitted plan. SHORTBURST is preferable for optimisation tasks where the goal is to find a better plan, not to certify the submitted plan.

4.3. Flip vs. BisectionEnsemble

BISECTIONENSEMBLE [Della-Libera, 2026c] runs short RECOM chains at each node of the bisection tree generated by CONVERGENCESWEEP. At each node, it samples alternative sub-district configurations that are compatible with the fixed higher-level splits. This makes BISECTIONENSEMBLE a *conditional* ensemble: it samples plans that share the same hierarchical structure as the submitted plan but vary in sub-district assignment.

The key difference from FLIP is the level of perturbation: BISECTIONENSEMBLE can change $O(n/k)$ tracts in a single node’s assignment, while FLIP changes exactly one tract globally. BISECTIONENSEMBLE is therefore better for characterising the set of plans with the same hierarchical structure (useful for demonstrating that the chosen structure is not uniquely favourable), while FLIP is better for certifying robustness to minimal perturbations.

4.4. Flip vs. ConvergenceSweep

CONVERGENCESWEEP [Della-Libera, 2026a] runs T independent METIS seeds and selects the best plan by edge-cut. The plans it generates are independent of each other and of the submitted plan; there is no notion of “neighbourhood” in the CONVERGENCESWEEP output.

FLIP, by contrast, generates plans that are within Hamming distance $\lfloor F \cdot p_{acc} \rfloor$ of the submitted plan in the worst case, though the actual Hamming distance of any visited plan is much smaller because the chain wanders randomly.

The two methods serve complementary purposes: CONVERGENCESWEEP answers “is there a better plan out there?” FLIP answers “is the submitted plan stable to local perturbations?” A complete submission uses both: CONVERGENCESWEEP to generate the plan, and FLIP to certify its local stability before filing.

4.5. Computational Cost in Practice

For North Carolina ($n = 2,200$, $k = 14$), typical wall-clock times on a single core are:

- FLIP ($F = 10,000$): approximately 1–3 seconds.
- SHORTBURST ($B = 50$ bursts, $L = 20$ steps): approximately 15–30 seconds.
- BISECTIONENSEMBLE (100 chains per node, 28 nodes): approximately 5–15 minutes.
- CONVERGENCESWEEP ($T = 600$ seeds): approximately 3–8 minutes.

FLIP is the cheapest method by a large margin. The full p -sweep (five FLIP runs at five percentiles) takes under 15 seconds and provides the legislative disclosure required by the audit protocol in Section 5.

5. The Flip Audit Chain

5.1. Manifest Record

Every FLIP run writes the following fields to the plan manifest (the JSON audit record that accompanies the submitted plan):

```
{
  "search": "flip",
  "base_seed": <u64>,
  "flip_steps": 10000,
  "percentile": 0.0,
  "visited_count": <usize>,
  "selected_plan_rank": <usize>,
  "chain_seed_0": <u64>
}
```

Here `visited_count` is `|visited|` (the number of distinct plans stored, equal to 1 plus the number of accepted steps), and `selected_plan_rank` is `[p · |visited|]` (the 0-indexed position of the selected plan in the EC-sorted visited list). The `chain_seed_0` field records the derived seed for deterministic reproduction.

5.2. Verification Protocol

A verifier who wishes to confirm that the submitted plan is the correct output of the stated FLIP run proceeds as follows:

1. Re-run the FLIP chain with the recorded `base_seed`, `flip_steps`, and `percentile` on the same census-tract graph and starting plan (which must also be recorded in the manifest).
2. Confirm that the re-run’s `visited_count` matches the manifest. A mismatch indicates either a different starting plan or a different adjacency graph was used.
3. Retrieve the plan at index `selected_plan_rank` in the EC-sorted visited list.
4. Compare this plan’s district assignment hash to the submitted plan’s hash. A match confirms the submitted plan is the correct FLIP output.

This protocol is stronger than seed-only recording in two ways. First, it confirms not just that the chain was run from a given seed, but that the specific plan selected (by p -percentile) matches. If an operator tried to substitute a different plan at the same percentile, the hash comparison would fail. Second, `visited_count` provides an independent check on the chain’s execution: if the acceptance rate differs substantially from the expected 50–65% range (implying fewer or more than 5,000–6,500 accepted steps out of 10,000), the verifier is alerted to investigate whether the tract adjacency graph or population data differs from the reference.

5.3. Why Seed-Only Recording Is Insufficient

Consider the following attack: an operator runs the FLIP chain, observes that the plan at $p = 0.5$ produces a favourable partisan outcome, and reports $p = 0.5$ in the manifest. Later, they claim they actually ran the chain with $p = 0.0$ and got the same plan “by coincidence.” Without `selected_plan_rank`, the verifier cannot distinguish these scenarios: both produce the same seed, and the seed alone does not determine which percentile was selected.

With `selected_plan_rank` recorded, the verifier re-runs the chain, inspects the plan at rank $\lfloor 0.0 \times |\text{visited}| \rfloor = 0$ (the minimum EC plan), and compares it to the submitted plan. If they differ, the reported percentile was wrong. This closes the manipulation window.

5.4. Integration with the SHA-256 Audit Chain

The `chain_seed_0` field is derived deterministically from the `base_seed` via the SHA-256 schedule:

$$\text{chain_seed_0} = \text{SHA256}\left(\text{"FLIP_CHAIN_"} \parallel 0^{\text{le64}} \parallel \text{"_"} \parallel \text{base_seed}^{\text{le64}}\right) [0:8].$$

Recording `chain_seed_0` explicitly lets the verifier confirm the seed derivation without re-implementing the SHA-256 schedule; they can simply check that `chain_seed_0` matches the expected derivation from `base_seed`.

The `base_seed` itself is part of the broader plan manifest SHA-256 chain [Della-Libera, 2026a], which links the FLIP run to the plan selection decision (which METIS seed produced the starting plan), the data fetch (which census-year shapefile was used), and the adjacency graph (which boundary-sharing rules were applied). This end-to-end chain makes it impossible to selectively manipulate any single step without breaking the chain.

6. Conclusion

6.1. Summary

The FLIP proposal is the simplest valid redistricting perturbation: move one boundary tract from its current district to an adjacent district, accept if the result is contiguous and population-balanced. Its properties follow directly from this simplicity:

Fast. Each step runs in $O(\text{deg})$ amortised time with incremental boundary set maintenance. Ten thousand steps take under 3 seconds for any state in our implementation.

High acceptance. Boundary tracts are near-balanced by selection; contiguity failure is rare. Acceptance rates of 40–70% are typical, giving 4,000–7,000 distinct visited plans per 10,000-step run.

Locally confined. Each FLIP step changes exactly one tract. The chain cannot tunnel through topological obstacles (Theorem 1), so it characterises only the local neighbourhood of the starting plan. This is a feature for sensitivity analysis and a bug for ensemble sampling.

Slow mixing. Moving from any starting plan to a fundamentally different plan requires $O(\text{district size})$ accepted steps at minimum (Theorem 3). In practice, the chain does not mix in 10,000 steps for any real-world state; it traces a random walk in the local neighbourhood.

6.2. Usage Guidelines

The FLIP chain should be used as follows in the redistricting workflow:

1. **After plan generation.** Generate the plan using CONVERGENCESWEEP (or another global search method). Do not use FLIP to generate the plan; use it only to certify the generated plan.
2. **Run the p -sweep.** Execute five FLIP runs at $p \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$. Record ρ_{seat} for each.
3. **Disclose the results.** Report `visited_count`, `selected_plan_rank`, σ_{EC} , and ρ_{seat} in the legislative findings. If any p -value produces a different seat outcome, explain the choice of percentile.
4. **Store the manifest.** Record `base_seed`, `chain_seed_0`, `visited_count`, and `selected_plan_rank` for verifier reproduction.

6.3. Open Questions

1. **Optimal step count.** Is $F = 10,000$ steps sufficient to characterise the local neighbourhood for all states? For large states ($n > 5,000$) or states with high district counts ($k > 20$), a larger F may be warranted. A stopping criterion based on the saturation of the visited-plan count would provide a principled answer.
2. **Weighting the acceptance.** The current FLIP implementation uses a hard-constraint acceptance criterion (accept if contiguous and balanced, reject otherwise). A soft Metropolis criterion that penalises large EC changes could concentrate the chain near the starting plan’s EC level, making the p -sweep more informative about plan quality.
3. **Multi-tract flip.** Allowing simultaneous moves of $r > 1$ adjacent boundary tracts would bridge topological obstacles and expand the reachable plan set, at the cost of

$O(r!)$ more proposals per step. This generalisation would interpolate between FLIP ($r = 1$) and RECOM ($r \approx n/k$).

4. **Formal partisan stability bounds.** Given a census-tract graph with known geometry (e.g., compact convex districts), can we derive a lower bound on ρ_{seat} under FLIP? Such a bound would let courts distinguish geometric constraint (high stability for topological reasons) from deliberate placement (artificially high stability due to gerrymandering).

6.4. Relation to the Full Toolkit

The redistricting toolkit is now complete at three perturbation levels. Table 1 (Section 4) summarises when to use each method. The standard workflow is:

CONVERGENCESWEEP (generate) $\rightarrow$ FLIP (certify) $\rightarrow$ BISECTIONENSEMBLE (structure
audit) $\rightarrow$ ensemble (global audit).

Each step adds a different type of evidence. FLIP occupies an essential niche: it is the only method that directly characterises the plan’s local neighbourhood at the tract level, and it is the only method fast enough to run interactively (under 10 seconds) during the plan-drafting process.

References

- Sarah Cannon, Ari Goldbloom-Helzner, Varun Gupta, Jesse Matthews, and Bhushan Suwal. Voting rights, markov chains, and optimization by short bursts. *arXiv*, 2022. arXiv:2011.02288. Introduces ShortBurst as a method for finding near-optimal redistricting plans. The burst length parameter controls the exploitation/exploration tradeoff, analogous to FLIP’s narrow local radius.
- Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f. ReCom proposal distribution. The primary chain whose global exploration radius is contrasted with the local radius of FLIP in Section 4.
- Gio Della-Libera. ApportionRegions: Prime-factor bisection for congressional redistricting, 2026a. B.11 in Algorithmic Redistricting series. 50-state sweep; 223D/209R national

headline. Defines the plan-level search methods (ConvergenceSweep, PercentileSweep) that FLIP complements at the local level.

Gio Della-Libera. Short-burst optimisation for redistricting: Short chains beat long chains, 2026b. G.6 in Algorithmic Redistricting series. Introduces ShortBurst as a trajectory-level search method. Provides the comparison baseline for Flip vs. ShortBurst in Section 4.

Gio Della-Libera. BisectionEnsemble: Node-level ensemble methods for structured redistricting, 2026c. H.1 in Algorithmic Redistricting series. Introduces node-level ReCom chains at each bisection tree node. Provides the intermediate perturbation level in the three-level hierarchy introduced in Section 1.

Alistair Sinclair. Improved bounds for mixing rates of Markov chains and multicommodity flow. *Combinatorics, Probability and Computing*, 1(4):351–370, 1992. doi: 10.1017/S0963548300000390. Multicommodity flow method for Markov chain mixing lower bounds. Establishes that local chains on combinatorial structures with polynomial diameter have exponential mixing times in the worst case. Supports the slow-mixing claim for single-tract flip chains in Section 2.

Supreme Court of the United States. *Rucho v. common cause*, 2019. 588 U.S. 684 (2019). Shifts partisan gerrymandering claims to state courts. Motivates local sensitivity analysis as a tool that state courts can deploy without requiring full ensemble methodology.